

RADIO MEASUREMENTS

A METHOD OF GENERATING ULTRA-SHORT RADIO PULSES

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A method of generating short nanosecond radio pulses from a sequence of long radio pulses is considered. It is shown that the proposed method may be used in the appropriate laboratory studies and tests of equipment intended for the reception and processing of wideband and ultra-wideband radio pulse signals. An algorithm and technique of generating short radio pulses are proposed. A detailed theoretical derivation of analytic expressions that describe the form of short radio pulses as well as a calculation of the permissible limits of the amplitude and phase errors of sinusoidal signals used to generate radio pulses are presented.

Keywords: radio pulse, ultra-short pulse, generator.

Appropriate generators of high-frequency signals are needed when performing tuning and regulation as well as testing of equipment used for reception and processing of wideband and ultra-wideband signals to see whether these signals conform to the technical conditions. Wideband and ultra-wideband signals are radiated by radar, radio navigation, radio communication, and other types of radio electronic equipment. For example, the use of acousto-optical meters of the parameters of radio signals is considered a promising means of estimating the parameters of such signals [1–3]. These meters implement different measuring algorithms [4, 5].

Generators of pico- and nano-second radio pulses are produced by a number of manufacturers of control and measuring devices, in particular, the firm of Keysight Technologies (United States), though because of their high cost they are not always available. Generators of radio pulse signals with pulse length amounting to several dozen or hundreds of nanoseconds and length of the front several nanoseconds are somewhat less expensive. Thus, the E8663D analog generator of high-frequency signals functions in the range of frequencies from 100 kHz to 9 GHz with resolution 0.001 Hz. The generator produces signals with amplitude, frequency, phase, and pulse modulation. Typical values of the parameters of a pulse with modulation by means of short pulses amount to 6 nsec of build-up (fall-off) time and length of 20 nsec in the range of carrier frequencies from 10 MHz to 9 GHz.

The possibility of generating a radar probe signal based on modern standard equipment from the firm of Rohde & Schwarz (Germany) is considered in [6, 7]. The equipment incorporates an ultra-high-frequency mixer, SMB100A generator of UHF signals, and AFQ100B generator of modulating signals. A high-frequency signal from an SMF100A generator enters one input of the mixer and a square waveform from an AFG100B generator of modulating signals is fed to the other input. In this design, the mixer plays the role of a multiplier at the output of which a sequence of radio pulses is generated. The cost of such an experimental setup is quite high.

In the present study, we will discuss the results of experimental investigations of a method of generating ultra-short radio pulses from relatively long radio pulses produced by a UHF signal generator. The carrier frequency of these signals is determined by the metrological characteristics of the particular generator. A flow chart of the method is presented in Fig.1. The operating principle consists in amplitude-phase processing of already existing long radio pulses, which are generated

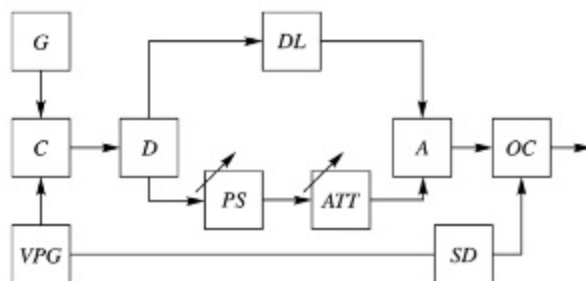


Fig. 1. Flow chart of generation of short radio pulses: *G* – generator of harmonic oscillations; *C* – commutator; *VPG* – video pulse generator; *D* – power divider; *DL* – delay line; *A* – adder; *PS* – phase shifter; *ATT* – attenuator; *OC* – output commutator; *SD* – strobe driver.

relatively simply by an ordinary commutation method. Either a UH generator with internal modulation or a harmonic oscillation generator *G* with external commutator *C* controlled by a video pulse generator *VPG* may serve as the source of the long pulses. The video pulse generator specifies the repetition period and length of the initial radio pulses. The length of these pulses may be arbitrarily great. It is limited only by the required repetition period of the initial pulse sequence. The initial sequence is divided by a power divider *D* into two channels, one of which contains a delay line *DL*. The delay time Δt determines the length of the radio pulses which have to be generated. The signals of the two channels are subsequently added by the adder *A* and, if necessary, amplified (the amplifier is not shown in the flow chart).

If conditions in which the signal enters the adder in antiphase and with identical amplitude are maintained, the signals mutually compensate each other in the region of their temporal crossing and remain unchanged outside this region. For this purpose, a phase shifter *PS* is installed in either channel and an attenuator *ATT* placed in the channel with greater amplitude. Ultimately, two short radio pulses are generated from the fronts and slumps of the initial signals.

The experimental mock-up developed by the present authors corresponds in general outline to the flow chart of Fig. 1. A generator from the firm of Agilent (now Keysight Technologies) with internal modulation serves as the source of the initial long radio pulses, an oscillogram of which is presented in Fig. 2a. A phase shifter with digital control checks the setting of the phase in 22.5° steps with error $\pm 5^\circ$. Strict antiphase of the signals cannot be achieved with such a significant step, hence fine phase adjustment was performed by continuous variation of the carrier frequency in a neighborhood of 1000 MHz.

The signal shown in Fig. 2b is generated at the output of the adder; the signals that intersect in time are compensated, forming short regions at the fronts and slumps of the initial sequence. Undisturbed (uncompensated) segments of the front of the initial pulse in one channel and slumps in the other channel constitute short radio pulses which follow at doubled repetition period relative to the initial sequence, which it is also necessary to generate. The structure of a radio pulse several nanoseconds in length generated by the method described here is shown in Fig. 2c in enlarged form. The radio pulse was obtained in the slump of the initial signal. The amplitude of the radio pulse is less than the initial pulse due to the use of only part of the slump. It should also be noted that the radio pulse is the more diffuse, the lengthier is the front (slump) of the initial sequence.

The experiment demonstrated that radio pulses of length significantly shorter not only at the vertex of the initial pulse, but also at the fronts, may be generated by this method. The basic step occurs on the segment of the front and the shorter is the initial front, the higher is the quality and the stronger, all other conditions being equal, is the radio pulse generated. The metrological properties of the generator in the range of carrier frequencies and the precision of the setting of the frequency of the generated signal remain unchanged and are determined by the parameters of the generator.

Let us analyze the process of generating short radio pulses. The instantaneous value of the initial long radio pulse at one of the inputs of the adder may be described by the expression

$$s_1(t) = A_1[h(t)(1 - e^{-t/\tau_c}) - h(t - \tau_0)(1 - e^{-(t-\tau_0)/\tau_c})]\cos(\omega_0 t + \varphi_1), \quad (1)$$

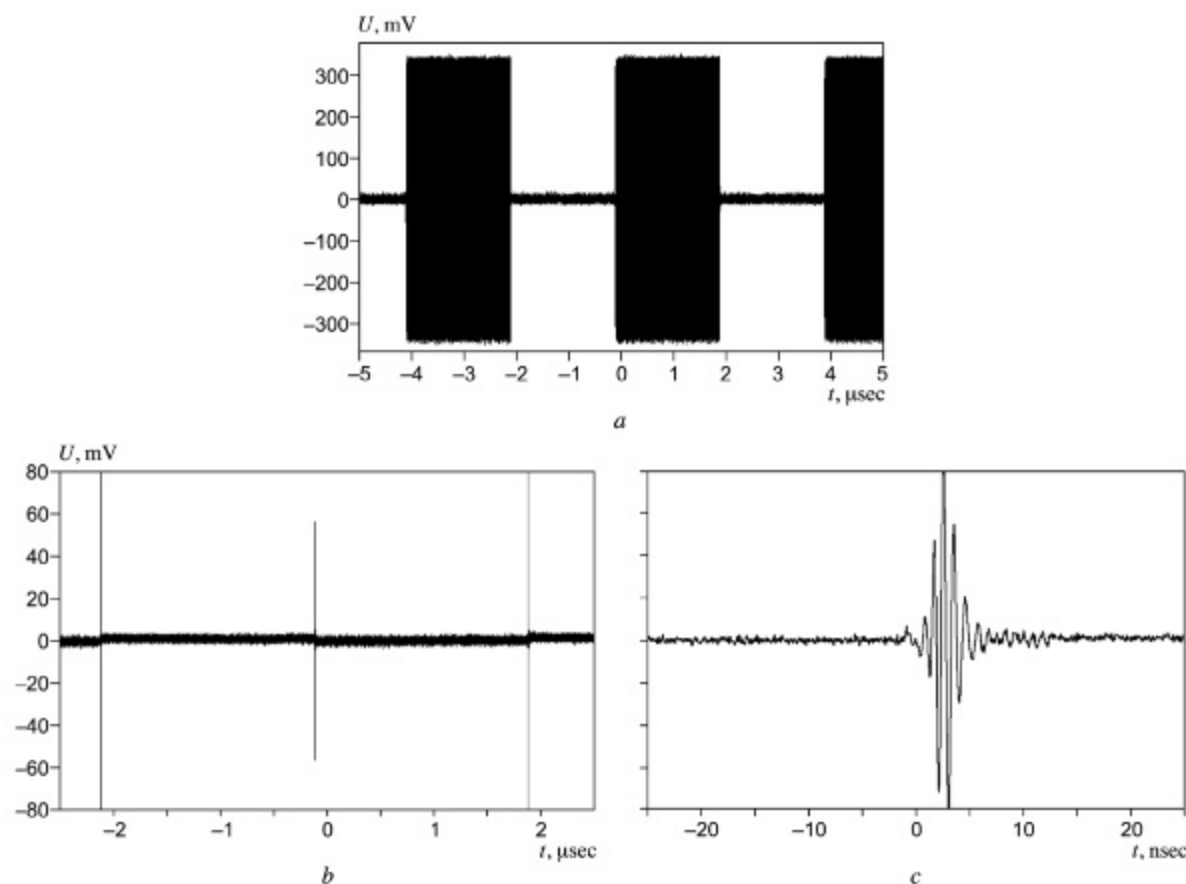


Fig. 2. Generation of short radio pulses: oscillogram of initial sequence (a); signal at output of adder (b); newly generated short radio pulse (c).

where A_1 is the amplitude of the radio pulse; $h(t)$, Heaviside function; t , time variable; τ_0 , length of radio pulse; ω_0 , circular frequency of carrier; τ_c , response time of commutator; and φ_1 , initial phase of signal.

The signal in the second input of the adder shifted by the delay line relative to $s_1(t)$ by a length of time equal to the required length of the short radio pulse $\tau = \Delta t$ may be described by analogy to (1), but shifted by τ :

$$s_2(t) = A_2[h(t - \tau)(1 - e^{-(t - \tau)/\tau_c}) - h(t - \tau - \tau_0)(1 - e^{-(t - \tau - \tau_0)/\tau_c})]\cos(\omega_0 t + \varphi_2). \quad (2)$$

At the output of the adder, where the conditions of equality of the amplitudes and antiphase of the signals in (1) and (2) are satisfied, i.e., when $A_1 = A_2$ and $\varphi_2 = \varphi_1 + \pi$, we have

$$s(t) = s_1(t) + s_2(t) = A_1[h(t)(1 - e^{-t/\tau_c}) - h(t - \tau)(1 - e^{-(t - \tau)/\tau_c}) - h(t - \tau_0)(1 - e^{-(t - \tau_0)/\tau_c}) + h(t - \tau - \tau_0)(1 - e^{-(t - \tau - \tau_0)/\tau_c})]\cos(\omega_0 t + \varphi_1). \quad (3)$$

In view of the fact that $(\tau, \tau_c) \ll \tau_0$, the latter expression represents two short radio pulses of length close to τ . The first pulse is generated by the first pair of terms in the brackets and the second by the last pair. There is no signal present in the interval between these pulses in the case of antiphase equal-amplitude addition. Where this condition is not satisfied, an

uncompensated (residual) spurious harmonic signal appears between the two useful signals. Let us determine the amplitude of the residual signal.

The signals in the inter-pulse interval may be described by ordinary harmonic functions of the same frequency:

$$S_1(t) = A_1 \cos(\omega_0 t - \varphi_1); \quad S_2(t) = A_2 \cos(\omega_0 t - \varphi_2).$$

Then, by [8], we may represent their sum as

$$S(t) = S_1(t) + S_2(t) = A \cos(\omega_0 t - \varphi),$$

where φ is the phase of the total oscillation and its amplitude

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_1 - \varphi_2)}. \quad (4)$$

Let us next introduce the relative error in the equality of the amplitudes of the initial signals:

$$\delta_A = (A_2 - A_1)/A_1,$$

as well as the absolute error of their antiphase:

$$\Delta_\varphi = \varphi_1 - \varphi_2 - \pi.$$

Then we may rewrite (4) in the form

$$A = A_1 \sqrt{1 + (1 + \delta_A)^2 + 2(1 + \delta_A) \cos(\pi + \Delta_\varphi)}.$$

Following some algebra, it assumes the form

$$A = A_1 \sqrt{4(1 + \delta_A) \sin^2(0.5 \Delta_\varphi) + \delta_A^2}. \quad (5)$$

From (5), it follows that in the case of ideally antiphase signals ($\Delta_\varphi = 0$), the amplitude of the spurious signal depends linearly on the amplitude error $A = A_1 \delta_A$, and if $\delta_A = 0$, this is also valid for the phase error:

$$A = 2A_1 \sin(0.5 \Delta_\varphi) = A_1 \Delta_\varphi.$$

Let us next estimate the magnitudes of the permissible errors δ_A and Δ_φ from the condition where the strength of the spurious signal is equal to some level P_N selected as the threshold level.

In light of (5), we may represent the strength of the spurious signal as

$$P_A = A^2 / (2R_0) = P_1 [4(1 + \delta_A) \sin^2(0.5 \Delta_\varphi) + \delta_A^2],$$

where $P_1 = A_1^2 / 2R_0$ is the strength of the signal at one of the inputs of the adder, and R_0 is the wave resistance of the line.

We obtain the condition

$$P_N / P_1 \geq 4(1 + \delta_A) \sin^2(0.5 \Delta_\varphi) + \delta_A^2, \quad (6)$$

which determines the region of permissible errors of the setting of the amplitudes of the signal and their antiphases.

In the general case, the region of permissible errors corresponds to the region bounded by the coordinate axes and the graph of the function

$$\Delta_\varphi(\delta_A) = 2 \arcsin \left[\sqrt{(P_N / P_1 - \delta_A^2) / (4(1 + \delta_A))} \right].$$

As an example, let us estimate the errors in the case where the threshold level P_N is the strength of the thermal noise [9]:

$$P_N = kT \Delta f, \quad (7)$$

where k is Boltzmann's constant; T , absolute temperature; and Δf , the transmission band of the former.

Thus, for $T = 295$ K, $P_1 = 1$ mW, and $\Delta f = 1$ GHz, we find from (6) and (7) the following estimate of the errors:

$$4(1 + \delta_A) \sin^2(0.5\Delta\varphi) + \delta_A^2 \leq 4 \cdot 10^{-9},$$

from which it follows that in order to achieve complete (within the framework of the given conditions) purification of the formed signal from the spurious signal, rather rigid constraints must be imposed on the phase shifter and attenuator of the former. Devices with continuous regulation must be used to implement these requirements.

To reduce the level of a spurious signal as well as eliminate an extra pulse (cf. Fig. 1), an additional output commutator OC controlled by a strobe driver SD may be introduced into the circuit. The requirements as to the speed of the commutator and former are low, since the length of the strobe may attain values practically equal to the repetition period.

It is necessary to specify the form of a function that approximates the front and sections of initial extended radio pulses to carry out a theoretical estimation of the precision of the setting of the length and amplitude of the newly constructed radio pulse. A host of variants for modeling the fronts at a given moment are used, and the selection of the most appropriate is possible only in the context of a particular device. With the use of the present method, it is possible to generate radio signals of nanosecond length by means of the minimum inventory of equipment available in any laboratory. Instead of high-frequency generators, less expensive synthesizers of harmonic signal may be used.

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